

Testing Documentation

Web-based Centralized Faculty Evaluation System

1. Master Test Plan

1.1 Purpose

The purpose of this Master Test Plan is to define the testing approach for the Web-based Centralized Faculty Evaluation System. It provides a structured procedure for verifying the functionality, reliability, and proper operation of the system before acceptance and actual use.

1.2 Testing Objectives

The testing aims to:

1. Verify that the system functions according to the identified requirements.
2. Identify and document system defects.
3. Verify that data is correctly processed and stored.
4. Ensure that different system modules work together properly.
5. Determine whether the system is ready for SAT and UAT.

1.3 Scope of Testing

Included

- User authentication
- User management
- Faculty management
- Faculty evaluation
- Evaluation submission
- Evaluation records
- Report generation
- Database operations
- User access/permissions

3. Testing Environment

Component	Tools
Operating System	
Frontend	Next.js, React, TypeScript, Tailwind CSS
Backend	Node.js, Express.js
Database	MySQL
Browser	Google Chrome
Development Tool	Visual Studio Code, Git, GitHub
Deployment	Vercel (Frontend), Railway (Backend and Database)

4. Unit Test Cases

ID	Module	Function	Test Input	Expected	Results	Status
001	Authentication	Log In	Valid Email and password	Redirect to Dashboard	Succesfull	Pass
002	Authentication	Log In	Incorrect password/user	Display error	Reject/error	Pass
003	Authentication	Log In	Empty Fields	Display Validation message	Reject/error	Pass
004	Authentication	Log In	Login with incomplete credentials	Email only / password only	Reject/Error	Pass
005	Authentication	Protected Page After Logout	Log out access previous dashboard	Prevents access redirect to landing page	Error	Fail
006	Authentication /Admin	Logout and Redirect	Click the Logout button	Redirect to the landing page	Error	Fail
007	Authentication	Invalid Login Attempts	Enter Incorrect Email/Pass	Appear Cooldown Period	Succesfull	Pass
008	Authentication	Token Persistence	Log In redirect, then reload	Authentication token remains stored after page reload	Error	Fail
009	Dashboard/School Head	Access Dashboard	Log In	User redirect to dashboard	Succesfull	Pass
0010	Dashboard/School Head	Open Management Tab	Click Management	Management content appears in the selected tab	Succesfull	Pass
0011	Dashboard/School Head	Open Forms Tab	Click Forms	Forms content appears in the selected tab	Succesfull	Pass
0012	Dashboard/School Head	Open Reports Tab	Click Reports	Reports content appears in the selected tab	Succesfull	Pass

0013	Dashboard/School Head	Open Teacher Tab	Click Teachers	Teacher content appears in the selected tab	Succesfull	Pass